

Q. 1 Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous, and suppose that $f(0) = f(1)$. Show that there is a point $a \in [0, \frac{2025}{2026}]$ satisfying $f(a) = f(a + \frac{1}{2026})$.

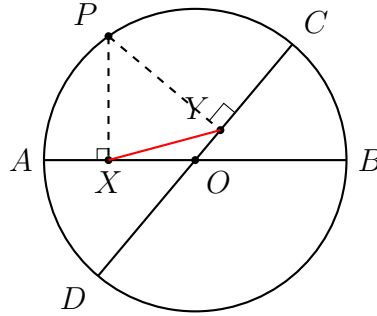
Q. 2 Given real numbers

$$0 = x_1 < x_2 < \cdots < x_{2026} < x_{2027} = 1,$$

with $x_{i+1} - x_i \leq \frac{1}{2}$ for $1 \leq i \leq 2026$. Show that

$$\frac{1}{4} < \sum_{i=1}^{1013} x_{2i}(x_{2i+1} - x_{2i-1}) < \frac{3}{4}.$$

Q. 3 Let AB and CD be two given diameters of a given circle with centre O . Let P be a point chosen arbitrarily on the circle. Drop perpendiculars from P to the two diameters, meeting them at X and Y . Prove that the length of XY does not change no matter where P is placed on the circle.



Q. 4 Let $a, b \in \mathbb{R}$ and $f(x) = a \cos x + b \sin 3x$. It is given that $f(x) > 1$ has no solutions in real. Prove that $|b| \leq 1$.

Q. 5 Let x_1, x_2 be the two roots of the equation $x^2 + px - 1 = 0$, where p is an odd integer. Let $y_n = x_1^n + x_2^n$, for $n \geq 0$. Show that y_n, y_{n+1} are coprime for all integers $n \geq 0$.

Q. 6 Let f be continuous and monotonically increasing, with $f(0) = 0$, and $f(1) = 1$. Prove that

$$\sum_{k=1}^9 f\left(\frac{k}{10}\right) + \sum_{k=1}^9 f^{-1}\left(\frac{k}{10}\right) \leq \frac{99}{10}.$$

Q. 7 Suppose by $[\cdot]$, we denote the greatest integer function. Answer the following:

- (a) For all $x, y \in \mathbb{R}$, show that $[2x] + [2y] \geq [x] + [y] + [x + y]$.
- (b) Using the previous result or otherwise, prove that

$$\frac{\binom{2m}{m} \binom{2n}{n}}{\binom{m+n}{n}}$$

is an integer for all $m, n \in \mathbb{N}$.

Q. 8 The length of each side of a convex quadrilateral $ABCD$ is strictly less than 24. Let P be any point inside $ABCD$. Prove that there is at least one vertex among A, B, C, D such that its distance from P is strictly less than 17.

-End of the Question Paper-